

Enhancing GD2 CAR T-cell therapy with IGF-1R blockade: Are DIPG CAR T cells ready for combinatorial therapy?

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The treatment landscape for patients with H3K27M-mutant diffuse midline glioma (DMG)/diffuse intrinsic pontine glioma (DIPG) is expanding in real time, with more investigators approaching this previously unsolvable disease at the mechanistic, translational, and clinical levels.¹ These efforts are starting to prove fruitful, with some durable responses seen in H3K27M-DMG patients with re-irradiation, the imipridone ONC201,² and most recently, chimeric antigen receptor (CAR) T clinical trials.^{3,4} The next phase of this translational effort will be to determine how to safely and effectively combine these therapies in novel combinations.

The recent publication by de Billy et al performs just this type of study, establishing a high-throughput screening to identify kinase inhibitors that enhance the efficacy of GD2 CAR T cells in H3K27M-DMG preclinical models.⁵ The group performed a platform of cell-based assays to screen 42 kinase inhibitors to determine the effect on H3K27M-DMG cells and CAR T-cell function. These studies pointed to the insulin-like growth factor 1/insulin receptor (IGF-1R/IR) antagonist linsitinib as being selectively toxic to H3K27M-DMG cells with minimal CAR T-cell exhaustion. GD2 CAR T cells plus linsitinib demonstrated combinatorial reduction of DIPG cell invasion into ex vivo pontine brain slices as well as reduced tumor growth in vivo.

GD2 expression was confirmed in pediatric high-grade glioma tumor tissue tested in this study, but variable in intensity and location (cytoplasmic and/or nuclear), and staining heterogeneity was independent of tumor location and molecular subgroup. When cell lines were derived from the hemispheric, H3-wildtype tumors, GD2 expression was lost.⁵ This loss of GD2 expression in vitro is an important reminder that no preclinical model is without imperfections and biomarkers are more likely to be truly “predictive” when they are measured within the tumor itself. Discounting the use of a GD2-targeted CAR T-cell therapy based only on cell line expression of GD2 may have inappropriately narrowed its potential use to only histone-mutant DMG.

Several GD2-directed CAR T-cell clinical trials have been or are being conducted in select pediatric cancers, including

neuroblastoma and pediatric DMG.³ In H3K27M-mutant DIPG/DMG, Majzner et al recently reported on the first four patients on their phase 1 study treated with GD2 CAR T-cell therapy, where three of four patients demonstrated radiographic and clinical benefit.³ The first dose of GD2 CAR T cells for each patient was given intravenously (IV), and subsequent doses given intracerebroventricularly (ICV) and toxicities were manageable with a well-developed supportive care algorithm. ICV administrations were associated with less systemic toxicities and increased pro-inflammatory cytokine presence in the cerebrospinal fluid (CSF) when compared with IV administration.³

Combination therapy is a cornerstone of cancer therapy and a logical next step in the continued development of CAR T-cell therapy. In neuroblastoma, combination strategies are being employed to enhance GD2 CAR T-cell activity, such as the co-administration of a MEK inhibitor⁶ or PD-1 inhibitor.⁷ In a high-throughput screening, de Billy et al found that BMS-754807 and linsitinib, both IGF-1R/IR inhibitors led to additive toxicity to GD2 CAR T-cell therapy without significant direct toxicity on the CAR T cells themselves.⁵ Linsitinib was also found to decrease the activation/exhaustion of the CAR T cells and an increase in central memory T cells.⁵

The IGF axis has been increasingly recognized to promote an immunosuppressive tumor microenvironment through negatively regulating dendritic cell activation, promoting regulatory T-cell proliferation, and inducing macrophage polarization to a pro-tumorigenic phenotype.⁸ Based on this, the IGF axis seemingly represents a reasonable target for targeted inhibition to promote a pro-inflammatory, anti-tumor environment. In this study, linsitinib in combination with GD2 CAR T cells were tested in vitro, and linsitinib was not required to cross the blood-brain barrier (BBB). Although to date, there are no published data describing linsitinib's ability to cross the BBB, utilization of the SwissADME tool⁹ to predict physicochemical properties of small molecules reveals that linsitinib is unlikely to cross the BBB. Future studies of this combination may tease out whether linsitinib is indeed entering the tumors (possibly

through CAR-T disruption of the BBB) or impacting the immune response/microenvironment through peripheral signaling.

Historically, the translation of therapies into the neuro-oncology clinic has covered the established track of in vitro cell culture to in vivo mouse tumor models to phase 1 study in humans. Unfortunately, successful preclinical studies based on just these two models have failed to produce effective therapies in DMG/DIPG. scRNA-seq data from H3K27M-tumors have confirmed that patient-derived xenografts (PDX) recapitulate the primary tumor better than in vitro models, and non-serum-containing neurosphere (NS) culture conditions are better than those with serum.¹⁰ While mouse models are useful, they lack the ability to assess the critical interaction between tumor cells and the microenvironment in real time. For invasive tumors like DIPG/DMG, this interaction is essential to the biology and growth of the tumor.

To this end, de Billy et al employ an ex vivo model in which human DIPG tumor cells implanted in the pontine region of mice brain are removed and cultured as an “organotypic brain slice” with or without treatment. This model allowed them to observe and quantify invasion. Optimization of this model, and the expanding use of induced pluripotent stem cell (iPSC) brain organoid models, should allow for time-lapsed videos of multiple fluorescently tagged tumors and normal cells to allow us to visualize and interrogate tumor cell invasion and response to therapy.

Studies like this that attempt to build on the early successes of a DMG/DIPG therapy demonstrate that we are indeed starting to finally break through the seemingly impenetrable armor of H3K27M-mutant DMG tumors. This is the time for us to press the gas pedal even further on multi-modal and combinatorial therapy for this patient population.

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